MCQ A · families of antiderivatives

After S53; MCQ B after S58.

For each question, only one option is correct. Justify your choice orally before checking the solutions.

1. An antiderivative of x^2 on $\mathbb{R}$ is:

A. $2x$

B. x^3

C. $x^3/3$

D. $3x^3$

2. All antiderivatives of e^{2x} are:

A. $e^{2x} + C$

B. $\frac{1}{2}e^{2x} + C$

C. $2e^{2x} + C$

D. $xe^{2x} + C$

3. An antiderivative of $2x/(1+x^2)$ on $\mathbb{R}$ is:

A. $\frac{1}{1+x^2}$

B. $2\ln x$

C. $\ln x$

D. $\ln(1+x^2)$

4. Two antiderivatives of the same function on an interval have graphs:

A. images of each other under a vertical translation

B. necessarily identical

C. always intersecting

D. images under any horizontal translati

5. The antiderivative of $2x$ satisfying $F(1) = 3$ is:

A. $x^2 + 3$ B. $2x^2 + 1$

C. $x^2 + 2$ D. $x^2 - 2$

MCQ B · differential equation and model

For each question, only one option is correct. Justify your choice orally before checking the solutions.

6. The solution of $y' = -2y$, $y(0) = 3$, is:

A. $3e^{2t}$ B. $3e^{-2t}$

C. $-2e^{3t}$ D. $3 - 2t$

7. The constant solution of $y' = -2y + 6$ is:

A. 6 B. -3

C. 2 D. 3

8. If y_p is a solution of $y' = ay + b(t)$, then $z = y - y_p$ satisfies:

A. $z' = az$ B. $z' = az + b(t)$

C. $z' = b(t)$ D. $z' = 0$ necessarily

9. Euler with step h for $y' = -y$, from $y_0 = 2$, gives:

A. $y_1 = 2 + h$ B. $y_1 = e^{-h}$

C. $y_1 = 2 - 2h$ D. $y_1 = 2 - h^2$

10. If t is in minutes and T in degrees Celsius, the coefficient k of $T' = -k(T - 20)$ is expressed in:

A. minutes B. reciprocal minutes

C. degrees Celsius D. litres

True or false · justification expected. For each statement, indicate whether it is true or false, then write a proof or a counterexample.

1. An antiderivative is unique without any further condition.

2. Euler's method gives the exact value for every differential equation.

3. Checking a solution consists of checking the equation and the initial condition.